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M2

|              |                                  |
|--------------|----------------------------------|
| Started on   | Sunday, 30 August 2026, 10:04 PM |
| State        | Finished                         |
| Completed on | Sunday, 30 August 2026, 10:05 PM |
| Time taken   | 38 secs                          |
| Marks        | 1.00/1.00                        |
| Grade        | 10.00 out of 10.00 (100%)        |

**Question 1** | Correct | Mark 1.00 out of 1.00

Given two arrays `array_One[]` and `array_Two[]` of same size `N`. We need to first rearrange the arrays such that the sum of the product of pairs (1 element from each) is minimum. That is  $\text{SUM}(A[i] * B[i])$  for all `i` is minimum.

**For example:**

| Input | Result |
|-------|--------|
| 3     | 28     |
| 1     |        |
| 2     |        |
| 3     |        |
| 4     |        |
| 5     |        |
| 6     |        |

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int ascending(const void *a, const void *b)
5  {
6      int x = *(const int *)a;
7      int y = *(const int *)b;
8
9      if (x < y) return -1;
10     if (x > y) return 1;
11     return 0;
12 }
13
14 int descending(const void *a, const void *b)
15 {
16     int x = *(const int *)a;
17     int y = *(const int *)b;
18
19     if (x < y) return 1;
20     if (x > y) return -1;
21     return 0;
22 }
23
24 int main()
25 {
26     int n;
27     scanf("%d", &n);
28
29     int a[n], b[n];
30
31     for (int i = 0; i < n; i++)
32         scanf("%d", &a[i]);
33
34     for (int i = 0; i < n; i++)
35         scanf("%d", &b[i]);
36
37     qsort(a, n, sizeof(int), ascending);
38     qsort(b, n, sizeof(int), descending);
39
40     long long sum = 0;
41
42     for (int i = 0; i < n; i++)
43         sum += (long long)a[i] * b[i];
44
45     printf("%lld", sum);
46
47     return 0;
48 }
```

|   | Input                                                     | Expected | Got |   |
|---|-----------------------------------------------------------|----------|-----|---|
| ✓ | 3<br>1<br>2<br>3<br>4<br>5<br>6                           | 28       | 28  | ✓ |
| ✓ | 4<br>7<br>5<br>1<br>2<br>1<br>3<br>4<br>1                 | 22       | 22  | ✓ |
| ✓ | 5<br>20<br>10<br>30<br>10<br>40<br>8<br>9<br>4<br>3<br>10 | 590      | 590 | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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